Supplementary development and temporal-confirmation results

Historical recalibration versus additional reference checks in CO sensor arrays

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Contents and numerical-file guide

S1: full-precision provenance and denominator definitions. S2: envelope conditions and contradiction handling. S3: complete RF family/pooled budget grid (Table S1; AGGREGATE_METRICS.csv). S4: limited OLS/ridge affine robustness (Table S2). S5: concentration support and relative tolerance (Table S3; METRICS_BY_RUN_FAMILY_BAND_SUBSET.csv). S6: timing (Table S4; QUERY_TIMING_BY_RUN.csv). S7: exact negative slope95 diagnostic (Table S5; SLOPE95_CROSSTAB.csv). S8: development-only conditional length sensitivity and all three day effects (Table S6; CONDITIONAL_RESAMPLING_EFFECTS.csv, LOSS_TIME_STRUCTURE.csv). S9: code/data access. S10: separately frozen fixed-D10 temporal confirmation, complete budget grids, family/session cells, containment and assurance. Every configuration/subset/count is in METRICS_BY_CONFIGURATION.csv. Figures 1–3 rebuild from SESSION_METRICS.csv and conditional effects. RESAMPLING_INDEX_MANIFEST.json identifies saved paired banks; EXCEEDANCE_SHARES.csv records weighted family numerators and denominators.

S1 Numerical provenance and denominators

Values derive from authentic saved development prediction arrays and causal replays. The local archive records a versioned SHA256 protocol/source manifest; no Git tag is claimed. Development historical calibration is D5/D7/D9, assessment D6/D8/D10, and all original fit/exclusion/reservation roles remain fixed. Each assessment has 100 shared exposures and 1,200 correlated configuration outputs, with six configurations per family. No reserved measurements enter these development results; the separately authorized temporal-confirmation evidence is reported in S10.

Original B0 copies are collapsed only after exact equality of predictions, references, query flags and intervals is checked. The original logs and removed-duplicate mapping remain intact. Three affine lambdas retain named identities but share one B0 numerical state. Every original bound/upstream sensitivity is retained in the CSV, even where only primary results are typeset. Each numerical row has complete model/bounds/schedule/budget/seed/day identity and source provenance; tables show four-decimal MAE and percent rates, while CSVs retain stored precision.

Random seeds 487–496 are averaged within each day before equal-day means. Seeds and overlapping peer configurations do not add physical replication. All/queried/unqueried supports and their own-subset consensus/B0 are retained. Empty queried B0 cells and unsupported bands

have undefined rates, not zero. The 10% relative sensitivity uses a strict exceedance and absolute 1 ppm at reference zero.

S2 Conditional envelope properties and contradiction fallback

The following properties assume every anchor inequality in the main text holds. They do not establish empirical future validity. All failed settings remain retained. If $\ell > u$, the raw model is contradictory. The original midpoint algorithm returns the midpoint of the descriptive hull,

$$[\ell^{\text{hull}}, u^{\text{hull}}] = \left[\min_{j \in \mathcal{J}_t}(e_j - w_{jt}),\ \max_{j \in \mathcal{J}_t}(e_j + w_{jt})\right], \tag{1}$$

while projection returns unchanged consensus. Assurance is withheld in either case. Hull containment does not restore the failed intersection model.

Conditional properties. If every anchor inequality holds, the actual residual belongs to the nonempty intersection and $|\widehat{y} - y| \leq R(c)$. The midpoint minimizes the largest absolute error over that interval because it equalizes the two endpoint distances. This conditional minimax property does not imply the smallest empirical MAE. Projection satisfies $|e - \text{clip}(0, \ell, u)| \leq |e|$ when $e \in [\ell, u]$: for a positive interval it subtracts ℓ, for a negative interval it subtracts u, and for a zero-containing interval it changes nothing. This is a standard interval-geometric fact. It does not apply outside the assumed interval.

A conditional assurance flag at tolerance τ requires a noncontradictory raw intersection and $R(c) \leq \tau$. It is not a calibrated probability. If a separate justified physical reference bound $|y - C| \leq u_{\text{ref}}$ were known, the triangle inequality would yield $|\widehat{y} - C| \leq R(c) + u_{\text{ref}}$. The released data do not establish those exposure-level hard bounds or their dependence. Reference-relative error sets must therefore not be identified with certified measurement uncertainty [16, 17].

S3 Complete primary RF results

Table S1: Complete primary RF all-exposure family and pooled grid. Rates in percent; changes from consensus at B0 and own B0 at positive budgets, in ppm. The three affine B0 states are identical. Ten random seeds averaged within each day. Full precision and exact counts reside in accompanying CSVs.

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE	Relative
ALL	Consensus	Historical only	0	0.8993	34.17	0.0000	34.17
ALL	Consensus	Periodic	5	0.8993	34.17	0.0000	34.17
ALL	Consensus	Random	5	0.8993	34.17	-0.0000	34.17
ALL	Consensus	Confidence	5	0.8993	34.17	0.0000	34.17
ALL	Consensus	Envelope	5	0.8993	34.17	0.0000	34.17
ALL	Consensus	Information	5	0.8993	34.17	0.0000	34.17
ALL	Consensus	Periodic	10	0.8993	34.17	0.0000	34.17
ALL	Consensus	Random	10	0.8993	34.17	-0.0000	34.17
ALL	Consensus	Confidence	10	0.8993	34.17	0.0000	34.17
ALL	Consensus	Envelope	10	0.8993	34.17	0.0000	34.17
ALL	Consensus	Information	10	0.8993	34.17	0.0000	34.17
ALL	Consensus	Periodic	20	0.8993	34.17	0.0000	34.17
ALL	Consensus	Random	20	0.8993	34.17	-0.0000	34.17
ALL	Consensus	Confidence	20	0.8993	34.17	0.0000	34.17
ALL	Consensus	Envelope	20	0.8993	34.17	0.0000	34.17
ALL	Consensus	Information	20	0.8993	34.17	0.0000	34.17
ALL	Midpoint	Historical only	0	0.9325	31.28	0.0332	33.53
ALL	Midpoint	Periodic	5	0.9304	32.06	-0.0021	34.25

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE	Relative
ALL	Midpoint	Random	5	0.9388	31.58	0.0063	33.60
ALL	Midpoint	Confidence	5	0.9545	32.28	0.0220	34.11
ALL	Midpoint	Envelope	5	0.9403	31.72	0.0077	33.92
ALL	Midpoint	Information	5	0.9040	30.36	-0.0285	32.36
ALL	Midpoint	Periodic	10	0.9717	32.06	0.0392	34.67
ALL	Midpoint	Random	10	0.9407	32.01	0.0082	33.87
ALL	Midpoint	Confidence	10	1.0122	33.75	0.0797	36.25
ALL	Midpoint	Envelope	10	0.9764	33.33	0.0439	34.11
ALL	Midpoint	Information	10	0.9356	31.81	0.0031	33.64
ALL	Midpoint	Periodic	20	0.9490	32.69	0.0165	34.11
ALL	Midpoint	Random	20	0.9431	32.27	0.0106	33.69
ALL	Midpoint	Confidence	20	0.9807	33.14	0.0482	35.28
ALL	Midpoint	Envelope	20	0.9418	31.86	0.0093	33.17
ALL	Midpoint	Information	20	0.9361	33.00	0.0036	33.78
ALL	Projection	Historical only	0	0.8942	33.97	-0.0052	34.00
ALL	Projection	Periodic	5	0.8942	33.97	0.0000	34.00
ALL	Projection	Random	5	0.8939	33.96	-0.0003	33.99
ALL	Projection	Confidence	5	0.8940	33.97	-0.0002	34.00
ALL	Projection	Envelope	5	0.8942	33.97	0.0000	34.00
ALL	Projection	Information	5	0.8942	33.97	0.0000	34.00
ALL	Projection	Periodic	10	0.8942	33.97	0.0000	34.00
ALL	Projection	Random	10	0.8937	33.94	-0.0005	33.98
ALL	Projection	Confidence	10	0.8942	33.94	0.0000	34.03
ALL	Projection	Envelope	10	0.8942	33.97	0.0000	34.00
ALL	Projection	Information	10	0.8935	33.92	-0.0007	33.94
ALL	Projection	Periodic	20	0.8930	33.83	-0.0012	33.94
ALL	Projection	Random	20	0.8932	33.92	-0.0010	33.96
ALL	Projection	Confidence	20	0.8941	33.94	-0.0001	34.00
ALL	Projection	Envelope	20	0.8941	33.97	-0.0001	34.00
ALL	Projection	Information	20	0.8935	33.92	-0.0007	33.94
ALL	Offset	Historical only	0	0.9000	33.78	0.0006	34.22
ALL	Offset	Periodic	5	0.8993	34.06	-0.0006	34.33
ALL	Offset	Random	5	0.8999	33.82	-0.0001	34.23
ALL	Offset	Confidence	5	0.9056	33.56	0.0056	34.14
ALL	Offset	Envelope	5	0.8993	33.92	-0.0007	34.28
ALL	Offset	Information	5	0.9000	33.78	-0.0000	34.33
ALL	Offset	Periodic	10	0.8995	33.94	-0.0005	34.31
ALL	Offset	Random	10	0.9005	33.93	0.0005	34.25
ALL	Offset	Confidence	10	0.9049	33.67	0.0049	34.42
ALL	Offset	Envelope	10	0.8993	33.81	-0.0006	34.28
ALL	Offset	Information	10	0.9002	33.86	0.0002	34.28
ALL	Offset	Periodic	20	0.9096	33.64	0.0096	34.53
ALL	Offset	Random	20	0.9010	33.86	0.0010	34.23
ALL	Offset	Confidence	20	0.9088	33.64	0.0088	34.33
ALL	Offset	Envelope	20	0.9004	34.00	0.0004	34.11
ALL	Offset	Information	20	0.9018	33.92	0.0019	34.33
ALL	Affine $\lambda = 1$	Historical only	0	0.7843	28.86	-0.1151	29.58
ALL	Affine $\lambda = 1$	Periodic	5	0.7835	28.81	-0.0007	29.64
ALL	Affine $\lambda = 1$	Random	5	0.7846	28.87	0.0004	29.47
ALL	Affine $\lambda = 1$	Confidence	5	0.7893	28.69	0.0050	29.31
ALL	Affine $\lambda = 1$	Envelope	5	0.7852	28.92	0.0009	29.36
ALL	Affine $\lambda = 1$	Information	5	0.7840	28.94	-0.0002	29.58
ALL	Affine $\lambda = 1$	Periodic	10	0.7843	28.83	0.0000	29.56
ALL	Affine $\lambda = 1$	Random	10	0.7857	28.83	0.0014	29.50
ALL	Affine $\lambda = 1$	Confidence	10	0.7921	28.94	0.0078	29.53
ALL	Affine $\lambda = 1$	Envelope	10	0.7848	29.00	0.0005	29.61
ALL	Affine $\lambda = 1$	Information	10	0.7857	29.03	0.0015	29.56

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE	Relative
ALL	Affine $\lambda = 1$	Periodic	20	0.7948	28.75	0.0105	29.47
ALL	Affine $\lambda = 1$	Random	20	0.7860	28.77	0.0017	29.41
ALL	Affine $\lambda = 1$	Confidence	20	0.7967	28.89	0.0125	29.47
ALL	Affine $\lambda = 1$	Envelope	20	0.7872	29.11	0.0029	29.67
ALL	Affine $\lambda = 1$	Information	20	0.7872	28.92	0.0030	29.89
ALL	Affine $\lambda = .95$	Periodic	5	0.7834	28.78	-0.0008	29.64
ALL	Affine $\lambda = .95$	Random	5	0.7847	28.85	0.0004	29.46
ALL	Affine $\lambda = .95$	Confidence	5	0.7900	28.67	0.0057	29.22
ALL	Affine $\lambda = .95$	Envelope	5	0.7854	28.86	0.0012	29.31
ALL	Affine $\lambda = .95$	Information	5	0.7840	28.97	-0.0003	29.58
ALL	Affine $\lambda = .95$	Periodic	10	0.7842	28.83	-0.0001	29.47
ALL	Affine $\lambda = .95$	Random	10	0.7860	28.80	0.0017	29.49
ALL	Affine $\lambda = .95$	Confidence	10	0.7938	28.97	0.0095	29.56
ALL	Affine $\lambda = .95$	Envelope	10	0.7850	28.92	0.0007	29.56
ALL	Affine $\lambda = .95$	Information	10	0.7868	29.06	0.0026	29.78
ALL	Affine $\lambda = .95$	Periodic	20	0.7989	28.67	0.0146	29.72
ALL	Affine $\lambda = .95$	Random	20	0.7867	28.73	0.0024	29.36
ALL	Affine $\lambda = .95$	Confidence	20	0.8038	29.11	0.0196	29.50
ALL	Affine $\lambda = .95$	Envelope	20	0.7899	29.06	0.0056	29.58
ALL	Affine $\lambda = .95$	Information	20	0.7913	29.11	0.0070	30.14
ALL	Affine $\lambda = .90$	Periodic	5	0.7833	28.78	-0.0010	29.58
ALL	Affine $\lambda = .90$	Random	5	0.7848	28.85	0.0005	29.47
ALL	Affine $\lambda = .90$	Confidence	5	0.7908	28.61	0.0066	29.31
ALL	Affine $\lambda = .90$	Envelope	5	0.7857	28.83	0.0015	29.36
ALL	Affine $\lambda = .90$	Information	5	0.7839	28.89	-0.0003	29.47
ALL	Affine $\lambda = .90$	Periodic	10	0.7841	28.81	-0.0001	29.64
ALL	Affine $\lambda = .90$	Random	10	0.7865	28.78	0.0022	29.44
ALL	Affine $\lambda = .90$	Confidence	10	0.7961	28.94	0.0119	29.61
ALL	Affine $\lambda = .90$	Envelope	10	0.7855	28.78	0.0012	29.56
ALL	Affine $\lambda = .90$	Information	10	0.7889	29.14	0.0046	29.86
ALL	Affine $\lambda = .90$	Periodic	20	0.8057	28.72	0.0215	29.89
ALL	Affine $\lambda = .90$	Random	20	0.7885	28.79	0.0042	29.35
ALL	Affine $\lambda = .90$	Confidence	20	0.8166	29.14	0.0323	29.92
ALL	Affine $\lambda = .90$	Envelope	20	0.7958	29.11	0.0115	30.17
ALL	Affine $\lambda = .90$	Information	20	0.8042	29.64	0.0200	30.81
Figaro	Consensus	Historical only	0	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Periodic	5	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Random	5	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Confidence	5	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Envelope	5	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Information	5	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Periodic	10	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Random	10	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Confidence	10	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Envelope	10	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Information	10	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Periodic	20	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Random	20	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Confidence	20	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Envelope	20	1.5289	58.67	0.0000	57.17
Figaro	Consensus	Information	20	1.5289	58.67	0.0000	57.17
Figaro	Midpoint	Historical only	0	1.5328	52.00	0.0038	52.28
Figaro	Midpoint	Periodic	5	1.5222	53.17	-0.0106	53.72
Figaro	Midpoint	Random	5	1.5455	52.49	0.0127	52.67
Figaro	Midpoint	Confidence	5	1.5543	52.61	0.0215	52.56
Figaro	Midpoint	Envelope	5	1.5532	52.67	0.0204	53.17
Figaro	Midpoint	Information	5	1.4813	50.17	-0.0514	50.83

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE	Relative
Figaro	Midpoint	Periodic	10	1.6171	53.22	0.0843	54.06
Figaro	Midpoint	Random	10	1.5382	52.62	0.0054	52.88
Figaro	Midpoint	Confidence	10	1.6376	54.17	0.1049	54.89
Figaro	Midpoint	Envelope	10	1.6333	55.89	0.1005	54.11
Figaro	Midpoint	Information	10	1.5388	52.78	0.0061	52.94
Figaro	Midpoint	Periodic	20	1.5327	51.67	-0.0001	52.94
Figaro	Midpoint	Random	20	1.5457	53.20	0.0130	53.23
Figaro	Midpoint	Confidence	20	1.5793	52.78	0.0465	53.61
Figaro	Midpoint	Envelope	20	1.5579	52.50	0.0252	53.22
Figaro	Midpoint	Information	20	1.5405	54.39	0.0077	53.44
Figaro	Projection	Historical only	0	1.5188	58.28	-0.0101	56.83
Figaro	Projection	Periodic	5	1.5188	58.28	0.0000	56.83
Figaro	Projection	Random	5	1.5183	58.24	-0.0005	56.82
Figaro	Projection	Confidence	5	1.5188	58.28	0.0000	56.83
Figaro	Projection	Envelope	5	1.5188	58.28	0.0000	56.83
Figaro	Projection	Information	5	1.5188	58.28	0.0000	56.83
Figaro	Projection	Periodic	10	1.5188	58.28	0.0000	56.83
Figaro	Projection	Random	10	1.5179	58.24	-0.0009	56.81
Figaro	Projection	Confidence	10	1.5189	58.22	0.0000	56.89
Figaro	Projection	Envelope	10	1.5188	58.28	0.0000	56.83
Figaro	Projection	Information	10	1.5174	58.17	-0.0015	56.72
Figaro	Projection	Periodic	20	1.5174	58.11	-0.0014	56.78
Figaro	Projection	Random	20	1.5171	58.19	-0.0018	56.77
Figaro	Projection	Confidence	20	1.5189	58.22	0.0000	56.89
Figaro	Projection	Envelope	20	1.5188	58.28	0.0000	56.83
Figaro	Projection	Information	20	1.5174	58.17	-0.0015	56.72
Figaro	Offset	Historical only	0	1.5038	57.94	-0.0251	57.17
Figaro	Offset	Periodic	5	1.5035	58.44	-0.0003	57.33
Figaro	Offset	Random	5	1.5037	58.03	-0.0001	57.22
Figaro	Offset	Confidence	5	1.5036	57.83	-0.0002	57.00
Figaro	Offset	Envelope	5	1.5038	58.17	-0.0000	57.39
Figaro	Offset	Information	5	1.5049	57.94	0.0011	57.44
Figaro	Offset	Periodic	10	1.5028	58.28	-0.0010	57.39
Figaro	Offset	Random	10	1.5038	58.31	-0.0000	57.25
Figaro	Offset	Confidence	10	1.5047	58.00	0.0008	57.39
Figaro	Offset	Envelope	10	1.5041	58.06	0.0002	57.28
Figaro	Offset	Information	10	1.5058	58.06	0.0019	57.33
Figaro	Offset	Periodic	20	1.5055	58.11	0.0017	57.67
Figaro	Offset	Random	20	1.5033	58.15	-0.0005	57.22
Figaro	Offset	Confidence	20	1.5063	57.89	0.0025	57.39
Figaro	Offset	Envelope	20	1.5054	58.22	0.0016	57.06
Figaro	Offset	Information	20	1.5067	58.28	0.0028	57.33
Figaro	Affine $\lambda = 1$	Historical only	0	1.2742	48.39	-0.2548	47.56
Figaro	Affine $\lambda = 1$	Periodic	5	1.2737	48.22	-0.0005	47.72
Figaro	Affine $\lambda = 1$	Random	5	1.2746	48.33	0.0004	47.60
Figaro	Affine $\lambda = 1$	Confidence	5	1.2728	48.44	-0.0014	47.22
Figaro	Affine $\lambda = 1$	Envelope	5	1.2772	48.44	0.0030	47.61
Figaro	Affine $\lambda = 1$	Information	5	1.2747	48.56	0.0006	47.67
Figaro	Affine $\lambda = 1$	Periodic	10	1.2738	48.28	-0.0003	47.61
Figaro	Affine $\lambda = 1$	Random	10	1.2754	48.27	0.0013	47.67
Figaro	Affine $\lambda = 1$	Confidence	10	1.2799	48.67	0.0057	47.67
Figaro	Affine $\lambda = 1$	Envelope	10	1.2767	48.50	0.0025	47.72
Figaro	Affine $\lambda = 1$	Information	10	1.2773	48.67	0.0031	48.11
Figaro	Affine $\lambda = 1$	Periodic	20	1.2778	48.39	0.0036	47.72
Figaro	Affine $\lambda = 1$	Random	20	1.2749	48.15	0.0007	47.49
Figaro	Affine $\lambda = 1$	Confidence	20	1.2832	48.67	0.0090	47.72
Figaro	Affine $\lambda = 1$	Envelope	20	1.2806	48.61	0.0064	48.22

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE	Relative
Figaro	Affine $\lambda = 1$	Information	20	1.2788	48.33	0.0047	48.56
Figaro	Affine $\lambda = .95$	Periodic	5	1.2737	48.17	-0.0005	47.78
Figaro	Affine $\lambda = .95$	Random	5	1.2746	48.31	0.0004	47.58
Figaro	Affine $\lambda = .95$	Confidence	5	1.2727	48.39	-0.0015	47.17
Figaro	Affine $\lambda = .95$	Envelope	5	1.2778	48.33	0.0036	47.56
Figaro	Affine $\lambda = .95$	Information	5	1.2748	48.56	0.0006	47.67
Figaro	Affine $\lambda = .95$	Periodic	10	1.2736	48.22	-0.0006	47.50
Figaro	Affine $\lambda = .95$	Random	10	1.2758	48.24	0.0016	47.66
Figaro	Affine $\lambda = .95$	Confidence	10	1.2822	48.83	0.0080	47.56
Figaro	Affine $\lambda = .95$	Envelope	10	1.2772	48.33	0.0030	47.72
Figaro	Affine $\lambda = .95$	Information	10	1.2793	48.67	0.0051	48.67
Figaro	Affine $\lambda = .95$	Periodic	20	1.2819	48.56	0.0077	48.11
Figaro	Affine $\lambda = .95$	Random	20	1.2756	48.08	0.0014	47.47
Figaro	Affine $\lambda = .95$	Confidence	20	1.2914	49.11	0.0172	48.11
Figaro	Affine $\lambda = .95$	Envelope	20	1.2847	48.50	0.0105	48.06
Figaro	Affine $\lambda = .95$	Information	20	1.2866	48.67	0.0125	49.33
Figaro	Affine $\lambda = .90$	Periodic	5	1.2736	48.17	-0.0006	47.67
Figaro	Affine $\lambda = .90$	Random	5	1.2746	48.30	0.0005	47.61
Figaro	Affine $\lambda = .90$	Confidence	5	1.2725	48.33	-0.0016	47.17
Figaro	Affine $\lambda = .90$	Envelope	5	1.2785	48.28	0.0043	47.67
Figaro	Affine $\lambda = .90$	Information	5	1.2748	48.39	0.0006	47.44
Figaro	Affine $\lambda = .90$	Periodic	10	1.2732	48.17	-0.0010	47.72
Figaro	Affine $\lambda = .90$	Random	10	1.2764	48.23	0.0022	47.64
Figaro	Affine $\lambda = .90$	Confidence	10	1.2854	48.78	0.0113	47.61
Figaro	Affine $\lambda = .90$	Envelope	10	1.2784	48.06	0.0042	47.89
Figaro	Affine $\lambda = .90$	Information	10	1.2831	48.78	0.0089	49.00
Figaro	Affine $\lambda = .90$	Periodic	20	1.2900	48.78	0.0158	48.22
Figaro	Affine $\lambda = .90$	Random	20	1.2780	48.26	0.0038	47.46
Figaro	Affine $\lambda = .90$	Confidence	20	1.3076	49.11	0.0334	48.28
Figaro	Affine $\lambda = .90$	Envelope	20	1.2938	48.61	0.0196	49.17
Figaro	Affine $\lambda = .90$	Information	20	1.3121	49.56	0.0379	50.50
FIS	Consensus	Historical only	0	0.2697	9.67	0.0000	11.17
FIS	Consensus	Periodic	5	0.2697	9.67	0.0000	11.17
FIS	Consensus	Random	5	0.2697	9.67	0.0000	11.17
FIS	Consensus	Confidence	5	0.2697	9.67	0.0000	11.17
FIS	Consensus	Envelope	5	0.2697	9.67	0.0000	11.17
FIS	Consensus	Information	5	0.2697	9.67	0.0000	11.17
FIS	Consensus	Periodic	10	0.2697	9.67	0.0000	11.17
FIS	Consensus	Random	10	0.2697	9.67	0.0000	11.17
FIS	Consensus	Confidence	10	0.2697	9.67	0.0000	11.17
FIS	Consensus	Envelope	10	0.2697	9.67	0.0000	11.17
FIS	Consensus	Information	10	0.2697	9.67	0.0000	11.17
FIS	Consensus	Periodic	20	0.2697	9.67	0.0000	11.17
FIS	Consensus	Random	20	0.2697	9.67	0.0000	11.17
FIS	Consensus	Confidence	20	0.2697	9.67	0.0000	11.17
FIS	Consensus	Envelope	20	0.2697	9.67	0.0000	11.17
FIS	Consensus	Information	20	0.2697	9.67	0.0000	11.17
FIS	Midpoint	Historical only	0	0.3322	10.56	0.0625	14.78
FIS	Midpoint	Periodic	5	0.3386	10.94	0.0063	14.78
FIS	Midpoint	Random	5	0.3321	10.68	-0.0002	14.53
FIS	Midpoint	Confidence	5	0.3546	11.94	0.0224	15.67
FIS	Midpoint	Envelope	5	0.3273	10.78	-0.0049	14.67
FIS	Midpoint	Information	5	0.3267	10.56	-0.0055	13.89
FIS	Midpoint	Periodic	10	0.3264	10.89	-0.0058	15.28
FIS	Midpoint	Random	10	0.3433	11.40	0.0111	14.86
FIS	Midpoint	Confidence	10	0.3869	13.33	0.0546	17.61
FIS	Midpoint	Envelope	10	0.3196	10.78	-0.0127	14.11

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE	Relative
FIS	Midpoint	Information	10	0.3325	10.83	0.0002	14.33
FIS	Midpoint	Periodic	20	0.3654	13.72	0.0332	15.28
FIS	Midpoint	Random	20	0.3405	11.34	0.0082	14.16
FIS	Midpoint	Confidence	20	0.3820	13.50	0.0498	16.94
FIS	Midpoint	Envelope	20	0.3257	11.22	-0.0065	13.11
FIS	Midpoint	Information	20	0.3317	11.61	-0.0006	14.11
FIS	Projection	Historical only	0	0.2696	9.67	-0.0002	11.17
FIS	Projection	Periodic	5	0.2696	9.67	0.0000	11.17
FIS	Projection	Random	5	0.2696	9.67	0.0000	11.17
FIS	Projection	Confidence	5	0.2692	9.67	-0.0003	11.17
FIS	Projection	Envelope	5	0.2696	9.67	0.0000	11.17
FIS	Projection	Information	5	0.2696	9.67	0.0000	11.17
FIS	Projection	Periodic	10	0.2696	9.67	0.0000	11.17
FIS	Projection	Random	10	0.2694	9.64	-0.0001	11.16
FIS	Projection	Confidence	10	0.2696	9.67	0.0001	11.17
FIS	Projection	Envelope	10	0.2696	9.67	0.0000	11.17
FIS	Projection	Information	10	0.2696	9.67	0.0000	11.17
FIS	Projection	Periodic	20	0.2687	9.56	-0.0009	11.11
FIS	Projection	Random	20	0.2694	9.64	-0.0002	11.16
FIS	Projection	Confidence	20	0.2693	9.67	-0.0002	11.11
FIS	Projection	Envelope	20	0.2694	9.67	-0.0002	11.17
FIS	Projection	Information	20	0.2696	9.67	0.0000	11.17
FIS	Offset	Historical only	0	0.2961	9.61	0.0264	11.28
FIS	Offset	Periodic	5	0.2952	9.67	-0.0009	11.33
FIS	Offset	Random	5	0.2960	9.61	-0.0001	11.24
FIS	Offset	Confidence	5	0.3076	9.28	0.0115	11.28
FIS	Offset	Envelope	5	0.2947	9.67	-0.0014	11.17
FIS	Offset	Information	5	0.2950	9.61	-0.0011	11.22
FIS	Offset	Periodic	10	0.2961	9.61	0.0000	11.22
FIS	Offset	Random	10	0.2972	9.56	0.0011	11.24
FIS	Offset	Confidence	10	0.3051	9.33	0.0089	11.44
FIS	Offset	Envelope	10	0.2946	9.56	-0.0015	11.28
FIS	Offset	Information	10	0.2946	9.67	-0.0016	11.22
FIS	Offset	Periodic	20	0.3136	9.17	0.0175	11.39
FIS	Offset	Random	20	0.2987	9.58	0.0025	11.23
FIS	Offset	Confidence	20	0.3113	9.39	0.0151	11.28
FIS	Offset	Envelope	20	0.2953	9.78	-0.0008	11.17
FIS	Offset	Information	20	0.2970	9.56	0.0009	11.33
FIS	Affine $\lambda = 1$	Historical only	0	0.2943	9.33	0.0246	11.61
FIS	Affine $\lambda = 1$	Periodic	5	0.2933	9.39	-0.0010	11.56
FIS	Affine $\lambda = 1$	Random	5	0.2946	9.41	0.0003	11.34
FIS	Affine $\lambda = 1$	Confidence	5	0.3058	8.94	0.0114	11.39
FIS	Affine $\lambda = 1$	Envelope	5	0.2932	9.39	-0.0012	11.11
FIS	Affine $\lambda = 1$	Information	5	0.2933	9.33	-0.0011	11.50
FIS	Affine $\lambda = 1$	Periodic	10	0.2947	9.39	0.0003	11.50
FIS	Affine $\lambda = 1$	Random	10	0.2959	9.39	0.0015	11.33
FIS	Affine $\lambda = 1$	Confidence	10	0.3042	9.22	0.0099	11.39
FIS	Affine $\lambda = 1$	Envelope	10	0.2929	9.50	-0.0014	11.50
FIS	Affine $\lambda = 1$	Information	10	0.2942	9.39	-0.0001	11.00
FIS	Affine $\lambda = 1$	Periodic	20	0.3117	9.11	0.0174	11.22
FIS	Affine $\lambda = 1$	Random	20	0.2971	9.38	0.0027	11.33
FIS	Affine $\lambda = 1$	Confidence	20	0.3103	9.11	0.0160	11.22
FIS	Affine $\lambda = 1$	Envelope	20	0.2938	9.61	-0.0006	11.11
FIS	Affine $\lambda = 1$	Information	20	0.2956	9.50	0.0012	11.22
FIS	Affine $\lambda = .95$	Periodic	5	0.2932	9.39	-0.0011	11.50
FIS	Affine $\lambda = .95$	Random	5	0.2948	9.39	0.0004	11.33
FIS	Affine $\lambda = .95$	Confidence	5	0.3072	8.94	0.0129	11.28

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE	Relative
FIS	Affine $\lambda = .95$	Envelope	5	0.2931	9.39	-0.0012	11.06
FIS	Affine $\lambda = .95$	Information	5	0.2932	9.39	-0.0012	11.50
FIS	Affine $\lambda = .95$	Periodic	10	0.2948	9.44	0.0005	11.44
FIS	Affine $\lambda = .95$	Random	10	0.2961	9.36	0.0018	11.32
FIS	Affine $\lambda = .95$	Confidence	10	0.3054	9.11	0.0111	11.56
FIS	Affine $\lambda = .95$	Envelope	10	0.2928	9.50	-0.0016	11.39
FIS	Affine $\lambda = .95$	Information	10	0.2944	9.44	0.0001	10.89
FIS	Affine $\lambda = .95$	Periodic	20	0.3158	8.78	0.0215	11.33
FIS	Affine $\lambda = .95$	Random	20	0.2978	9.38	0.0034	11.25
FIS	Affine $\lambda = .95$	Confidence	20	0.3163	9.11	0.0219	10.89
FIS	Affine $\lambda = .95$	Envelope	20	0.2951	9.61	0.0007	11.11
FIS	Affine $\lambda = .95$	Information	20	0.2959	9.56	0.0016	10.94
FIS	Affine $\lambda = .90$	Periodic	5	0.2930	9.39	-0.0013	11.50
FIS	Affine $\lambda = .90$	Random	5	0.2949	9.39	0.0006	11.33
FIS	Affine $\lambda = .90$	Confidence	5	0.3091	8.89	0.0147	11.44
FIS	Affine $\lambda = .90$	Envelope	5	0.2930	9.39	-0.0013	11.06
FIS	Affine $\lambda = .90$	Information	5	0.2930	9.39	-0.0013	11.50
FIS	Affine $\lambda = .90$	Periodic	10	0.2950	9.44	0.0007	11.56
FIS	Affine $\lambda = .90$	Random	10	0.2965	9.33	0.0022	11.25
FIS	Affine $\lambda = .90$	Confidence	10	0.3068	9.11	0.0125	11.61
FIS	Affine $\lambda = .90$	Envelope	10	0.2925	9.50	-0.0018	11.22
FIS	Affine $\lambda = .90$	Information	10	0.2947	9.50	0.0003	10.72
FIS	Affine $\lambda = .90$	Periodic	20	0.3215	8.67	0.0272	11.56
FIS	Affine $\lambda = .90$	Random	20	0.2990	9.32	0.0047	11.25
FIS	Affine $\lambda = .90$	Confidence	20	0.3256	9.17	0.0312	11.56
FIS	Affine $\lambda = .90$	Envelope	20	0.2978	9.61	0.0035	11.17
FIS	Affine $\lambda = .90$	Information	20	0.2963	9.72	0.0020	11.11

S4 Limited upstream-model sensitivity

RF remains primary. No sensor model is refitted. OLS/ridge reuse their own saved observables and masks; confidence is peer MAD rather than RF tree spread. Only consensus and affine factors 1/.95 at B0/B20 are added to the previous midpoint sensitivity. This is not an outcome-selected model competition.

Table S2: M9: saved OLS/ridge inputs, unchanged consensus and inverse affine factors 1/.95 at B0/B20. Model-specific confidence and envelope masks retained. Common affine B0 shown once.

Model	Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE
OLS	ALL	Consensus	Historical only	0	0.8652	32.25	0.0000
OLS	ALL	Consensus	Periodic	20	0.8652	32.25	0.0000
OLS	ALL	Consensus	Random	20	0.8652	32.25	0.0000
OLS	ALL	Consensus	Confidence	20	0.8652	32.25	0.0000
OLS	ALL	Consensus	Envelope	20	0.8652	32.25	0.0000
OLS	ALL	Consensus	Information	20	0.8652	32.25	0.0000
OLS	ALL	Affine $\lambda = 1$	Historical only	0	0.7678	27.58	-0.0974
OLS	ALL	Affine $\lambda = 1$	Periodic	20	0.7663	27.47	-0.0015
OLS	ALL	Affine $\lambda = 1$	Random	20	0.7649	27.46	-0.0029
OLS	ALL	Affine $\lambda = 1$	Confidence	20	0.7693	28.11	0.0015
OLS	ALL	Affine $\lambda = 1$	Envelope	20	0.7657	27.47	-0.0021
OLS	ALL	Affine $\lambda = 1$	Information	20	0.7642	27.17	-0.0036
OLS	ALL	Affine $\lambda = .95$	Periodic	20	0.7655	27.33	-0.0024
OLS	ALL	Affine $\lambda = .95$	Random	20	0.7639	27.37	-0.0039
OLS	ALL	Affine $\lambda = .95$	Confidence	20	0.7704	27.92	0.0026
OLS	ALL	Affine $\lambda = .95$	Envelope	20	0.7668	27.72	-0.0011

Model	Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE
OLS	ALL	Affine $\lambda = .95$	Information	20	0.7641	27.36	-0.0037
OLS	Figaro	Consensus	Historical only	0	1.2713	50.33	0.0000
OLS	Figaro	Consensus	Periodic	20	1.2713	50.33	0.0000
OLS	Figaro	Consensus	Random	20	1.2713	50.33	-0.0000
OLS	Figaro	Consensus	Confidence	20	1.2713	50.33	0.0000
OLS	Figaro	Consensus	Envelope	20	1.2713	50.33	0.0000
OLS	Figaro	Consensus	Information	20	1.2713	50.33	0.0000
OLS	Figaro	Affine $\lambda = 1$	Historical only	0	1.0865	43.00	-0.1848
OLS	Figaro	Affine $\lambda = 1$	Periodic	20	1.0823	43.28	-0.0042
OLS	Figaro	Affine $\lambda = 1$	Random	20	1.0811	42.93	-0.0055
OLS	Figaro	Affine $\lambda = 1$	Confidence	20	1.0900	44.11	0.0034
OLS	Figaro	Affine $\lambda = 1$	Envelope	20	1.0815	43.17	-0.0051
OLS	Figaro	Affine $\lambda = 1$	Information	20	1.0804	42.44	-0.0061
OLS	Figaro	Affine $\lambda = .95$	Periodic	20	1.0809	43.11	-0.0056
OLS	Figaro	Affine $\lambda = .95$	Random	20	1.0793	42.82	-0.0072
OLS	Figaro	Affine $\lambda = .95$	Confidence	20	1.0914	44.00	0.0049
OLS	Figaro	Affine $\lambda = .95$	Envelope	20	1.0819	43.50	-0.0046
OLS	Figaro	Affine $\lambda = .95$	Information	20	1.0808	42.72	-0.0058
OLS	FIS	Consensus	Historical only	0	0.4591	14.17	0.0000
OLS	FIS	Consensus	Periodic	20	0.4591	14.17	0.0000
OLS	FIS	Consensus	Random	20	0.4591	14.17	-0.0000
OLS	FIS	Consensus	Confidence	20	0.4591	14.17	0.0000
OLS	FIS	Consensus	Envelope	20	0.4591	14.17	0.0000
OLS	FIS	Consensus	Information	20	0.4591	14.17	0.0000
OLS	FIS	Affine $\lambda = 1$	Historical only	0	0.4491	12.17	-0.0099
OLS	FIS	Affine $\lambda = 1$	Periodic	20	0.4504	11.67	0.0012
OLS	FIS	Affine $\lambda = 1$	Random	20	0.4488	11.99	-0.0003
OLS	FIS	Affine $\lambda = 1$	Confidence	20	0.4487	12.11	-0.0005
OLS	FIS	Affine $\lambda = 1$	Envelope	20	0.4499	11.78	0.0008
OLS	FIS	Affine $\lambda = 1$	Information	20	0.4480	11.89	-0.0011
OLS	FIS	Affine $\lambda = .95$	Periodic	20	0.4500	11.56	0.0009
OLS	FIS	Affine $\lambda = .95$	Random	20	0.4484	11.92	-0.0007
OLS	FIS	Affine $\lambda = .95$	Confidence	20	0.4494	11.83	0.0003
OLS	FIS	Affine $\lambda = .95$	Envelope	20	0.4516	11.94	0.0025
OLS	FIS	Affine $\lambda = .95$	Information	20	0.4475	12.00	-0.0016
Ridge	ALL	Consensus	Historical only	0	1.0873	39.08	0.0000
Ridge	ALL	Consensus	Periodic	20	1.0873	39.08	0.0000
Ridge	ALL	Consensus	Random	20	1.0873	39.08	-0.0000
Ridge	ALL	Consensus	Confidence	20	1.0873	39.08	0.0000
Ridge	ALL	Consensus	Envelope	20	1.0873	39.08	0.0000
Ridge	ALL	Consensus	Information	20	1.0873	39.08	0.0000
Ridge	ALL	Affine $\lambda = 1$	Historical only	0	0.9046	34.36	-0.1826
Ridge	ALL	Affine $\lambda = 1$	Periodic	20	0.9071	34.00	0.0025
Ridge	ALL	Affine $\lambda = 1$	Random	20	0.9047	34.17	0.0001
Ridge	ALL	Affine $\lambda = 1$	Confidence	20	0.9075	34.28	0.0028
Ridge	ALL	Affine $\lambda = 1$	Envelope	20	0.9021	34.22	-0.0025
Ridge	ALL	Affine $\lambda = 1$	Information	20	0.9071	34.06	0.0025
Ridge	ALL	Affine $\lambda = .95$	Periodic	20	0.9077	34.06	0.0031
Ridge	ALL	Affine $\lambda = .95$	Random	20	0.9043	34.20	-0.0003
Ridge	ALL	Affine $\lambda = .95$	Confidence	20	0.9085	34.17	0.0039
Ridge	ALL	Affine $\lambda = .95$	Envelope	20	0.9018	34.53	-0.0028
Ridge	ALL	Affine $\lambda = .95$	Information	20	0.9085	34.08	0.0039
Ridge	Figaro	Consensus	Historical only	0	1.5425	54.67	0.0000
Ridge	Figaro	Consensus	Periodic	20	1.5425	54.67	0.0000
Ridge	Figaro	Consensus	Random	20	1.5425	54.67	0.0000
Ridge	Figaro	Consensus	Confidence	20	1.5425	54.67	0.0000
Ridge	Figaro	Consensus	Envelope	20	1.5425	54.67	0.0000

Model	Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE
Ridge	Figaro	Consensus	Information	20	1.5425	54.67	0.0000
Ridge	Figaro	Affine $\lambda = 1$	Historical only	0	1.2159	48.67	-0.3266
Ridge	Figaro	Affine $\lambda = 1$	Periodic	20	1.2154	48.67	-0.0005
Ridge	Figaro	Affine $\lambda = 1$	Random	20	1.2150	48.71	-0.0009
Ridge	Figaro	Affine $\lambda = 1$	Confidence	20	1.2195	48.78	0.0036
Ridge	Figaro	Affine $\lambda = 1$	Envelope	20	1.2099	48.44	-0.0060
Ridge	Figaro	Affine $\lambda = 1$	Information	20	1.2179	48.67	0.0020
Ridge	Figaro	Affine $\lambda = .95$	Periodic	20	1.2164	48.67	0.0005
Ridge	Figaro	Affine $\lambda = .95$	Random	20	1.2144	48.81	-0.0015
Ridge	Figaro	Affine $\lambda = .95$	Confidence	20	1.2214	48.83	0.0055
Ridge	Figaro	Affine $\lambda = .95$	Envelope	20	1.2091	48.67	-0.0068
Ridge	Figaro	Affine $\lambda = .95$	Information	20	1.2191	48.83	0.0033
Ridge	FIS	Consensus	Historical only	0	0.6320	23.50	0.0000
Ridge	FIS	Consensus	Periodic	20	0.6320	23.50	0.0000
Ridge	FIS	Consensus	Random	20	0.6320	23.50	-0.0000
Ridge	FIS	Consensus	Confidence	20	0.6320	23.50	0.0000
Ridge	FIS	Consensus	Envelope	20	0.6320	23.50	0.0000
Ridge	FIS	Consensus	Information	20	0.6320	23.50	0.0000
Ridge	FIS	Affine $\lambda = 1$	Historical only	0	0.5934	20.06	-0.0387
Ridge	FIS	Affine $\lambda = 1$	Periodic	20	0.5988	19.33	0.0054
Ridge	FIS	Affine $\lambda = 1$	Random	20	0.5943	19.64	0.0010
Ridge	FIS	Affine $\lambda = 1$	Confidence	20	0.5955	19.78	0.0021
Ridge	FIS	Affine $\lambda = 1$	Envelope	20	0.5944	20.00	0.0010
Ridge	FIS	Affine $\lambda = 1$	Information	20	0.5964	19.44	0.0030
Ridge	FIS	Affine $\lambda = .95$	Periodic	20	0.5991	19.44	0.0057
Ridge	FIS	Affine $\lambda = .95$	Random	20	0.5943	19.59	0.0009
Ridge	FIS	Affine $\lambda = .95$	Confidence	20	0.5955	19.50	0.0022
Ridge	FIS	Affine $\lambda = .95$	Envelope	20	0.5946	20.39	0.0012
Ridge	FIS	Affine $\lambda = .95$	Information	20	0.5979	19.33	0.0045

S5 Family and concentration support

Table S3: RF affine B0 concentration support. Boundaries are evaluation-only; the relative research tolerance is 10% when reference is positive and 1 ppm at zero. All arms/subsets retain exact support in the CSV.

Day	Family	Band	Exposures	Outputs	MAE	P_1	Relative (%)
D6	Figaro	[0,5]	30	180	1.2350	32.22	55.56
D6	Figaro	(5,10]	20	120	1.4842	45.83	52.50
D6	Figaro	(10,20]	50	300	1.5110	61.67	43.33
D6	FIS	[0,5]	30	180	0.2378	5.00	21.11
D6	FIS	(5,10]	20	120	0.3686	3.33	15.00
D6	FIS	(10,20]	50	300	0.2794	10.00	6.00
D8	Figaro	[0,5]	30	180	0.8498	36.67	60.56
D8	Figaro	(5,10]	20	120	1.0776	48.33	61.67
D8	Figaro	(10,20]	50	300	1.2868	46.67	30.33
D8	FIS	[0,5]	30	180	0.1498	1.67	10.00
D8	FIS	(5,10]	20	120	0.2591	10.00	10.00
D8	FIS	(10,20]	50	300	0.2359	7.33	2.00
D10	Figaro	[0,5]	30	180	1.0910	45.56	64.44
D10	Figaro	(5,10]	20	120	1.2639	45.00	53.33
D10	Figaro	(10,20]	50	300	1.4114	57.67	36.33
D10	FIS	[0,5]	30	180	0.2682	5.56	15.56
D10	FIS	(5,10]	20	120	0.5466	22.50	25.00

Day	Family	Band	Exposures	Outputs	MAE	P_1	Relative (%)
D10	FIS	(10,20]	50	300	0.3876	17.00	13.67

S6 Paid query timing

Table S4: Paid query timing, mean per session after random-seed averaging. Only the final query has no later scored prediction under day reset; no purchase is refunded. Per-run queries remain in the archive.

Model	Schedule	B	Last 10	At final
RF	Periodic	5	1.00	0.00
RF	Random	5	0.80	0.00
RF	Confidence	5	1.00	0.33
RF	Envelope	5	0.00	0.00
RF	Information	5	0.33	0.00
RF	Periodic	10	1.00	0.00
RF	Random	10	0.70	0.00
RF	Confidence	10	1.00	0.33
RF	Envelope	10	0.00	0.00
RF	Information	10	0.00	0.00
RF	Periodic	20	2.00	0.00
RF	Random	20	2.30	0.20
RF	Confidence	20	1.33	0.33
RF	Envelope	20	1.00	0.33
RF	Information	20	0.00	0.00
OLS	Periodic	20	2.00	0.00
OLS	Random	20	2.30	0.20
OLS	Confidence	20	0.33	0.00
OLS	Envelope	20	0.67	0.00
OLS	Information	20	0.00	0.00
Ridge	Periodic	20	2.00	0.00
Ridge	Random	20	2.30	0.20
Ridge	Confidence	20	1.00	0.00
Ridge	Envelope	20	1.33	0.00
Ridge	Information	20	0.00	0.00

S7 Contradiction, containment and assurance

Table S5: Complete RF slope95/envelope B20 cross-tab, 3,600 outputs. All zero cells retained. Reported hull is the original numerical fallback on contradictions. Exact identities are saved.

Contradiction	Raw outside	Reported outside	Assured	Count
0	0	0	0	429
0	0	0	1	1099
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	135
0	1	1	1	1188
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	749
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

False assurance is computed independently as a noncontradictory, own-center radius flag and an actual absolute prediction-minus-reference error strictly above the same tolerance. At 1 ppm, RF slope95 envelope B20 midpoint has 503/2,287 false/assured (21.99%) and 503/3,600 false/all (13.97%); primary has 0/60 (0%) and 0/3,600 (0%). The 1,188 outside-and-assured entries describe interval noncontainment, not actual errors above 1 ppm. FALSE_ASSURANCE_AUDIT.csv retains all saved available interval centers, model/bounds identities, budgets, seeds, days, families and all/queried/unqueried subsets at 0.5/1/2 ppm, with undefined denominators marked NA. Offset and affine assurance remains unavailable. Noncontainment is a separate column.

S8 Development-only conditional resampling and temporal structure

No confirmation bootstrap was performed. For development D6/D8/D10 only, paired overlapping blocks use 2,000 replicates, seed 487 and fixed lengths 5/10/20. One index bank per length/day is shared across all arms/families/endpoints. Configurations are not resampled independently. Fixed fits, trajectories, initialization and shared reference labels are conditioned upon; learning and reference uncertainty are not resimulated. Percentile intervals are conditional sensitivity bounds, requiring regularity assumptions for an inferential interpretation. Adaptive chronological loss paths may be nonstationary. Neither a zero crossing nor an excluded zero establishes equivalence, causal drift explanation or independent-array coverage. The complete CSV retains every comparator/metric, all three session effects and direction counts. Quarter means, lag-one correlations and linear trends inspect temporal structure without validating arbitrary nonstationarity.

Table S6: Development-only conditional resampling sensitivity, B20 envelope/information schedules. All three session changes and block lengths retained; the complete endpoint/comparator grid is in the accompanying conditional-resampling CSV. Bounds condition on original fits, paths and references.

Scope	Estimator	Schedule	L	I MAE	Bounds	D6	D8	D10
ALL	Affine $\lambda = 1$	Envelope	5	0.0029	[0.0003, 0.0061]	0.0065	0.0015	0.0007
ALL	Affine $\lambda = 1$	Envelope	10	0.0029	[-0.0001, 0.0056]	0.0065	0.0015	0.0007
ALL	Affine $\lambda = 1$	Envelope	20	0.0029	[-0.0003, 0.0050]	0.0065	0.0015	0.0007

Scope	Estimator	Schedule	L	I MAE	Bounds	D6	D8	D10
ALL	Affine $\lambda = 1$	Information	5	0.0030	[-0.0007, 0.0069]	0.0055	0.0008	0.0026
ALL	Affine $\lambda = 1$	Information	10	0.0030	[-0.0007, 0.0065]	0.0055	0.0008	0.0026
ALL	Affine $\lambda = 1$	Information	20	0.0030	[-0.0010, 0.0054]	0.0055	0.0008	0.0026
ALL	Affine $\lambda = .95$	Envelope	5	0.0056	[0.0007, 0.0113]	0.0122	0.0028	0.0019
ALL	Affine $\lambda = .95$	Envelope	10	0.0056	[-0.0001, 0.0109]	0.0122	0.0028	0.0019
ALL	Affine $\lambda = .95$	Envelope	20	0.0056	[-0.0005, 0.0097]	0.0122	0.0028	0.0019
ALL	Affine $\lambda = .95$	Information	5	0.0070	[0.0009, 0.0136]	0.0156	0.0027	0.0027
ALL	Affine $\lambda = .95$	Information	10	0.0070	[0.0005, 0.0129]	0.0156	0.0027	0.0027
ALL	Affine $\lambda = .95$	Information	20	0.0070	[0.0002, 0.0115]	0.0156	0.0027	0.0027
ALL	Affine $\lambda = .90$	Envelope	5	0.0115	[0.0023, 0.0220]	0.0242	0.0050	0.0053
ALL	Affine $\lambda = .90$	Envelope	10	0.0115	[0.0008, 0.0212]	0.0242	0.0050	0.0053
ALL	Affine $\lambda = .90$	Envelope	20	0.0115	[0.0000, 0.0194]	0.0242	0.0050	0.0053
ALL	Affine $\lambda = .90$	Information	5	0.0200	[0.0074, 0.0336]	0.0502	0.0060	0.0037
ALL	Affine $\lambda = .90$	Information	10	0.0200	[0.0056, 0.0337]	0.0502	0.0060	0.0037
ALL	Affine $\lambda = .90$	Information	20	0.0200	[0.0037, 0.0330]	0.0502	0.0060	0.0037
Figaro	Affine $\lambda = 1$	Envelope	5	0.0064	[0.0020, 0.0118]	0.0111	0.0040	0.0040
Figaro	Affine $\lambda = 1$	Envelope	10	0.0064	[0.0016, 0.0106]	0.0111	0.0040	0.0040
Figaro	Affine $\lambda = 1$	Envelope	20	0.0064	[0.0019, 0.0102]	0.0111	0.0040	0.0040
Figaro	Affine $\lambda = 1$	Information	5	0.0047	[-0.0018, 0.0115]	0.0101	0.0052	-0.0013
Figaro	Affine $\lambda = 1$	Information	10	0.0047	[-0.0018, 0.0104]	0.0101	0.0052	-0.0013
Figaro	Affine $\lambda = 1$	Information	20	0.0047	[-0.0013, 0.0085]	0.0101	0.0052	-0.0013
Figaro	Affine $\lambda = .95$	Envelope	5	0.0105	[0.0024, 0.0200]	0.0204	0.0056	0.0055
Figaro	Affine $\lambda = .95$	Envelope	10	0.0105	[0.0017, 0.0184]	0.0204	0.0056	0.0055
Figaro	Affine $\lambda = .95$	Envelope	20	0.0105	[0.0016, 0.0173]	0.0204	0.0056	0.0055
Figaro	Affine $\lambda = .95$	Information	5	0.0125	[0.0006, 0.0248]	0.0293	0.0090	-0.0009
Figaro	Affine $\lambda = .95$	Information	10	0.0125	[0.0005, 0.0232]	0.0293	0.0090	-0.0009
Figaro	Affine $\lambda = .95$	Information	20	0.0125	[0.0011, 0.0207]	0.0293	0.0090	-0.0009
Figaro	Affine $\lambda = .90$	Envelope	5	0.0196	[0.0041, 0.0366]	0.0404	0.0083	0.0100
Figaro	Affine $\lambda = .90$	Envelope	10	0.0196	[0.0026, 0.0341]	0.0404	0.0083	0.0100
Figaro	Affine $\lambda = .90$	Envelope	20	0.0196	[0.0019, 0.0321]	0.0404	0.0083	0.0100
Figaro	Affine $\lambda = .90$	Information	5	0.0379	[0.0139, 0.0639]	0.0964	0.0151	0.0023
Figaro	Affine $\lambda = .90$	Information	10	0.0379	[0.0103, 0.0640]	0.0964	0.0151	0.0023
Figaro	Affine $\lambda = .90$	Information	20	0.0379	[0.0064, 0.0637]	0.0964	0.0151	0.0023
FIS	Affine $\lambda = 1$	Envelope	5	-0.0006	[-0.0025, 0.0014]	0.0018	-0.0010	-0.0026
FIS	Affine $\lambda = 1$	Envelope	10	-0.0006	[-0.0030, 0.0016]	0.0018	-0.0010	-0.0026
FIS	Affine $\lambda = 1$	Envelope	20	-0.0006	[-0.0037, 0.0007]	0.0018	-0.0010	-0.0026
FIS	Affine $\lambda = 1$	Information	5	0.0012	[-0.0006, 0.0035]	0.0009	-0.0035	0.0064
FIS	Affine $\lambda = 1$	Information	10	0.0012	[-0.0010, 0.0036]	0.0009	-0.0035	0.0064
FIS	Affine $\lambda = 1$	Information	20	0.0012	[-0.0013, 0.0032]	0.0009	-0.0035	0.0064
FIS	Affine $\lambda = .95$	Envelope	5	0.0007	[-0.0024, 0.0042]	0.0040	-0.0000	-0.0018
FIS	Affine $\lambda = .95$	Envelope	10	0.0007	[-0.0031, 0.0045]	0.0040	-0.0000	-0.0018
FIS	Affine $\lambda = .95$	Envelope	20	0.0007	[-0.0039, 0.0029]	0.0040	-0.0000	-0.0018
FIS	Affine $\lambda = .95$	Information	5	0.0016	[-0.0008, 0.0044]	0.0020	-0.0035	0.0062
FIS	Affine $\lambda = .95$	Information	10	0.0016	[-0.0014, 0.0047]	0.0020	-0.0035	0.0062
FIS	Affine $\lambda = .95$	Information	20	0.0016	[-0.0019, 0.0042]	0.0020	-0.0035	0.0062
FIS	Affine $\lambda = .90$	Envelope	5	0.0035	[-0.0018, 0.0096]	0.0080	0.0018	0.0006
FIS	Affine $\lambda = .90$	Envelope	10	0.0035	[-0.0028, 0.0102]	0.0080	0.0018	0.0006
FIS	Affine $\lambda = .90$	Envelope	20	0.0035	[-0.0038, 0.0076]	0.0080	0.0018	0.0006
FIS	Affine $\lambda = .90$	Information	5	0.0020	[-0.0017, 0.0061]	0.0040	-0.0031	0.0051
FIS	Affine $\lambda = .90$	Information	10	0.0020	[-0.0024, 0.0070]	0.0040	-0.0031	0.0051
FIS	Affine $\lambda = .90$	Information	20	0.0020	[-0.0032, 0.0060]	0.0040	-0.0031	0.0051

S9 Code and data access

The original UCI measurements are public under DOI 10.24432/C5S302. The complete permitted local development archive contains pre-query exports, source hashes, model/mask identities, causal

logs, original training/replay code and derived development feature inputs. All tables and figures are supplied as static LaTeX build inputs alongside unchanged full-precision numerical files. The document-only figure utility reads those saved summary values to improve presentation; it performs no model fitting, prediction replay, metric recomputation or resampling. A public code/prediction deposit is not yet available; the final access and licensing arrangement awaits author approval. The temporal confirmation uses later sessions on the same apparatus. No reserved raw inputs are distributed.

S10 Fixed-D10 temporal confirmation

These are separately authorized final D11/D12/D13 results, not additional development folds. Upstream RF fitting uses D1,D2,D4–D9 only; correction calibration and controller normalization use D10 only. Every session, policy, budget and seed resets to that same initialized state. R7/R14 and UCI270 batches 7–10 remain excluded. All 129 final paths were committed before summaries, and saved-output recomputation reproduces all 40,635 run metric rows and 5,040 aggregates. The protocol/source/model hashes and development evidence remain unchanged; no refit, retuning or final-result method selection occurred.

The original pooled primary contrasts are historical affine $\lambda = 1$ B0 minus raw consensus and incremental affine $\lambda = 1$ envelope B20 minus own B0. The frozen descriptive criterion requires lower strict 1 ppm exceedance and nonpositive MAE change pooled and separately in both families; strict exceedance is primary and MAE is the guardrail. Historical benefit is heterogeneous because FIS MAE worsens on all three days. Incremental benefit meets the pooled-and-both-family aggregate criterion, but session reversals preclude uniform improvement. Envelope superiority over all three original schedule comparators is unsupported. FIS MAE improves relative to affine B0 (0.9471 to 0.9033 ppm) but remains worse than unchanged consensus (0.8820 ppm). The full cell tables retain all unfavorable values.

Table S7: All primary confirmation day/family MAE cells (ppm). H is affine B0 minus raw consensus; I is cumulative affine envelope B20 minus own B0. Positive changes are adverse.

Day	Scope	Raw	Affine B0	Affine B20	H	I
D11	ALL	2.1576	2.1302	2.0384	-0.0274	-0.0918
D11	Figaro	2.5710	2.3961	2.3533	-0.1750	-0.0427
D11	FIS	1.7441	1.8644	1.7234	0.1202	-0.1409
D12	ALL	1.0316	0.8921	0.8970	-0.1395	0.0049
D12	Figaro	1.5836	1.2794	1.2865	-0.3042	0.0071
D12	FIS	0.4796	0.5048	0.5075	0.0252	0.0027
D13	ALL	1.1818	1.0437	1.0417	-0.1380	-0.0021
D13	Figaro	1.9413	1.6153	1.6044	-0.3260	-0.0109
D13	FIS	0.4222	0.4721	0.4789	0.0499	0.0068

Table S8: Exact primary confirmation strict 1 ppm exceedance counts. N counts outputs; each day has 100 shared exposures.

Day	Scope	N	Raw	Affine B0	Affine B20
D11	ALL	1200	774	683	662
D11	Figaro	600	453	359	356
D11	FIS	600	321	324	306
D12	ALL	1200	477	414	410
D12	Figaro	600	372	312	308
D12	FIS	600	105	102	102
D13	ALL	1200	540	453	445
D13	Figaro	600	450	364	355

Day	Scope	N	Raw	Affine B0	Affine B20
D13	FIS	600	90	89	90

Table S9: All fifteen prespecified pooled affine B20 confirmation contrasts. MAE changes in ppm; exceedance changes in percentage points. Random seeds are averaged within day before equal-day aggregation.

Factor	Schedule	I MAE	I P_1 (pp)	D11	D12	D13
$\lambda = 1$	Periodic	-0.0154	-0.2222	-0.0736	0.0089	0.0185
$\lambda = 1$	Random	-0.0254	-0.4889	-0.0743	-0.0008	-0.0011
$\lambda = 1$	Confidence	-0.0375	0.4444	-0.1179	0.0010	0.0044
$\lambda = 1$	Envelope	-0.0297	-0.9167	-0.0918	0.0049	-0.0021
$\lambda = 1$	Information	-0.0464	-1.0833	-0.1248	-0.0029	-0.0114
$\lambda = .95$	Periodic	-0.0153	-0.0556	-0.0822	0.0118	0.0245
$\lambda = .95$	Random	-0.0275	-0.4389	-0.0813	-0.0006	-0.0006
$\lambda = .95$	Confidence	-0.0245	-0.1944	-0.0903	0.0050	0.0118
$\lambda = .95$	Envelope	-0.0316	-0.9722	-0.0991	0.0080	-0.0037
$\lambda = .95$	Information	-0.0551	-1.4722	-0.1486	-0.0030	-0.0139
$\lambda = .90$	Periodic	-0.0128	0.0556	-0.0903	0.0161	0.0359
$\lambda = .90$	Random	-0.0286	-0.3694	-0.0869	0.0001	0.0011
$\lambda = .90$	Confidence	0.0047	1.8056	-0.0299	0.0171	0.0270
$\lambda = .90$	Envelope	-0.0314	-1.0833	-0.1016	0.0142	-0.0070
$\lambda = .90$	Information	-0.0606	-1.7778	-0.1654	0.0004	-0.0169

At primary envelope B20, midpoint raw/reported noncontainment counts are 284/276 out of 3,600, with eight contradictions. D11 gives 263/255, eight contradictions and 37/55 false/assured; D12 gives 5/5, zero contradictions and 0/12; D13 gives 16/16, zero contradictions and 1/7. Each session has 1,200 outputs. Family false/assured counts are Figaro 1/1 and FIS 37/73. Overall false assurance is 38/74 (51.35%) conditional and 38/3,600 (1.06%) joint. Queried counts are 8/14 and 8/720; unqueried are 30/60 and 30/2,880. Historical B0 midpoint has 242 raw/reported noncontainments and 3/3 false assurances; its Figaro conditional rate is undefined because none are assured. These are midpoint-envelope diagnostics, not affine uncertainty intervals. Assurance uses the midpoint's own radius and actual error, not interval noncontainment alone; affine/offset assurance remains unavailable.

The all/queried/unqueried numerical records retain 0.5/1/2 ppm endpoints, RMSE, bias, own-B0 and same-subset consensus, exact supports, per-configuration values, timing and fixed-seed ranges. Undefined queried B0 denominators are NA. Seeds are averaged within day, then days equally; counts sum seed-mean day numerators, and RMSE uses squared-error sums. The 300 shared final exposures and 3,600 correlated outputs are distinct from the 300 development exposures. Computational paths/purchases do not add physical replication. No final conditional resampling analysis is part of this frozen executable scope.

Table S10: Complete prespecified confirmation grid on all exposures. Rates in percent. Changes are from raw consensus at B0 and own B0 at positive budgets; all three affine B0 states are identical. Full-precision session/seed/configuration/subset records are in C5_METRICS.csv and C5_AGGREGATE.csv; empty queried B0 cells are NA.

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE
ALL	Consensus	Historical only	0	1.4570	49.75	0.0000
ALL	Consensus	Periodic	5	1.4570	49.75	0.0000
ALL	Consensus	Random	5	1.4570	49.75	0.0000
ALL	Consensus	Confidence	5	1.4570	49.75	0.0000
ALL	Consensus	Envelope	5	1.4570	49.75	0.0000

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE
ALL	Consensus	Information	5	1.4570	49.75	0.0000
ALL	Consensus	Periodic	10	1.4570	49.75	0.0000
ALL	Consensus	Random	10	1.4570	49.75	0.0000
ALL	Consensus	Confidence	10	1.4570	49.75	0.0000
ALL	Consensus	Envelope	10	1.4570	49.75	0.0000
ALL	Consensus	Information	10	1.4570	49.75	0.0000
ALL	Consensus	Periodic	20	1.4570	49.75	0.0000
ALL	Consensus	Random	20	1.4570	49.75	0.0000
ALL	Consensus	Confidence	20	1.4570	49.75	0.0000
ALL	Consensus	Envelope	20	1.4570	49.75	0.0000
ALL	Consensus	Information	20	1.4570	49.75	0.0000
ALL	Midpoint	Historical only	0	1.5571	47.47	0.1002
ALL	Midpoint	Periodic	5	1.5396	46.56	-0.0175
ALL	Midpoint	Random	5	1.5679	47.66	0.0107
ALL	Midpoint	Confidence	5	1.5198	48.36	-0.0373
ALL	Midpoint	Envelope	5	1.6040	48.28	0.0469
ALL	Midpoint	Information	5	1.5185	46.97	-0.0386
ALL	Midpoint	Periodic	10	1.5760	49.50	0.0189
ALL	Midpoint	Random	10	1.5810	48.30	0.0238
ALL	Midpoint	Confidence	10	1.5285	48.58	-0.0286
ALL	Midpoint	Envelope	10	1.5700	47.14	0.0129
ALL	Midpoint	Information	10	1.5004	46.11	-0.0568
ALL	Midpoint	Periodic	20	1.6167	49.97	0.0596
ALL	Midpoint	Random	20	1.5541	48.37	-0.0031
ALL	Midpoint	Confidence	20	1.5133	49.00	-0.0439
ALL	Midpoint	Envelope	20	1.5383	47.08	-0.0188
ALL	Midpoint	Information	20	1.4879	48.31	-0.0692
ALL	Projection	Historical only	0	1.4569	49.75	-0.0001
ALL	Projection	Periodic	5	1.4537	49.61	-0.0032
ALL	Projection	Random	5	1.4556	49.69	-0.0013
ALL	Projection	Confidence	5	1.4524	50.22	-0.0045
ALL	Projection	Envelope	5	1.4563	49.75	-0.0006
ALL	Projection	Information	5	1.4528	49.53	-0.0041
ALL	Projection	Periodic	10	1.4501	49.92	-0.0068
ALL	Projection	Random	10	1.4553	49.78	-0.0016
ALL	Projection	Confidence	10	1.4450	50.22	-0.0119
ALL	Projection	Envelope	10	1.4562	49.75	-0.0007
ALL	Projection	Information	10	1.4512	49.53	-0.0057
ALL	Projection	Periodic	20	1.4541	49.97	-0.0027
ALL	Projection	Random	20	1.4495	49.79	-0.0074
ALL	Projection	Confidence	20	1.4425	50.28	-0.0144
ALL	Projection	Envelope	20	1.4494	49.86	-0.0075
ALL	Projection	Information	20	1.4455	49.72	-0.0114
ALL	Offset	Historical only	0	1.4602	49.22	0.0032
ALL	Offset	Periodic	5	1.4451	49.06	-0.0151
ALL	Offset	Random	5	1.4531	49.05	-0.0071
ALL	Offset	Confidence	5	1.4233	48.56	-0.0369
ALL	Offset	Envelope	5	1.4519	48.97	-0.0083
ALL	Offset	Information	5	1.4514	49.14	-0.0088
ALL	Offset	Periodic	10	1.4333	48.56	-0.0269
ALL	Offset	Random	10	1.4451	48.94	-0.0151
ALL	Offset	Confidence	10	1.4103	47.83	-0.0499
ALL	Offset	Envelope	10	1.4480	48.75	-0.0122
ALL	Offset	Information	10	1.4455	49.14	-0.0147
ALL	Offset	Periodic	20	1.4442	48.72	-0.0160
ALL	Offset	Random	20	1.4349	48.61	-0.0253
ALL	Offset	Confidence	20	1.3993	47.17	-0.0609

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE
ALL	Offset	Envelope	20	1.4281	48.47	-0.0321
ALL	Offset	Information	20	1.4253	48.67	-0.0349
ALL	Affine $\lambda = 1$	Historical only	0	1.3553	43.06	-0.1016
ALL	Affine $\lambda = 1$	Periodic	5	1.3442	42.67	-0.0112
ALL	Affine $\lambda = 1$	Random	5	1.3479	42.81	-0.0074
ALL	Affine $\lambda = 1$	Confidence	5	1.3269	43.17	-0.0284
ALL	Affine $\lambda = 1$	Envelope	5	1.3448	42.47	-0.0106
ALL	Affine $\lambda = 1$	Information	5	1.3444	43.06	-0.0110
ALL	Affine $\lambda = 1$	Periodic	10	1.3289	42.61	-0.0264
ALL	Affine $\lambda = 1$	Random	10	1.3399	42.69	-0.0154
ALL	Affine $\lambda = 1$	Confidence	10	1.3167	43.44	-0.0386
ALL	Affine $\lambda = 1$	Envelope	10	1.3415	42.14	-0.0139
ALL	Affine $\lambda = 1$	Information	10	1.3362	42.61	-0.0192
ALL	Affine $\lambda = 1$	Periodic	20	1.3399	42.83	-0.0154
ALL	Affine $\lambda = 1$	Random	20	1.3299	42.57	-0.0254
ALL	Affine $\lambda = 1$	Confidence	20	1.3179	43.50	-0.0375
ALL	Affine $\lambda = 1$	Envelope	20	1.3257	42.14	-0.0297
ALL	Affine $\lambda = 1$	Information	20	1.3090	41.97	-0.0464
ALL	Affine $\lambda = .95$	Historical only	0	1.3553	43.06	-0.1016
ALL	Affine $\lambda = .95$	Periodic	5	1.3436	42.64	-0.0118
ALL	Affine $\lambda = .95$	Random	5	1.3474	42.79	-0.0079
ALL	Affine $\lambda = .95$	Confidence	5	1.3259	43.25	-0.0294
ALL	Affine $\lambda = .95$	Envelope	5	1.3438	42.44	-0.0115
ALL	Affine $\lambda = .95$	Information	5	1.3437	43.06	-0.0117
ALL	Affine $\lambda = .95$	Periodic	10	1.3269	42.56	-0.0285
ALL	Affine $\lambda = .95$	Random	10	1.3387	42.66	-0.0167
ALL	Affine $\lambda = .95$	Confidence	10	1.3158	43.53	-0.0396
ALL	Affine $\lambda = .95$	Envelope	10	1.3402	42.06	-0.0151
ALL	Affine $\lambda = .95$	Information	10	1.3352	42.61	-0.0202
ALL	Affine $\lambda = .95$	Periodic	20	1.3400	43.00	-0.0153
ALL	Affine $\lambda = .95$	Random	20	1.3278	42.62	-0.0275
ALL	Affine $\lambda = .95$	Confidence	20	1.3308	42.86	-0.0245
ALL	Affine $\lambda = .95$	Envelope	20	1.3238	42.08	-0.0316
ALL	Affine $\lambda = .95$	Information	20	1.3002	41.58	-0.0551
ALL	Affine $\lambda = .90$	Historical only	0	1.3553	43.06	-0.1016
ALL	Affine $\lambda = .90$	Periodic	5	1.3429	42.58	-0.0125
ALL	Affine $\lambda = .90$	Random	5	1.3469	42.77	-0.0085
ALL	Affine $\lambda = .90$	Confidence	5	1.3252	43.44	-0.0302
ALL	Affine $\lambda = .90$	Envelope	5	1.3428	42.36	-0.0125
ALL	Affine $\lambda = .90$	Information	5	1.3429	42.97	-0.0124
ALL	Affine $\lambda = .90$	Periodic	10	1.3245	42.50	-0.0308
ALL	Affine $\lambda = .90$	Random	10	1.3374	42.72	-0.0179
ALL	Affine $\lambda = .90$	Confidence	10	1.3166	43.61	-0.0387
ALL	Affine $\lambda = .90$	Envelope	10	1.3394	42.19	-0.0160
ALL	Affine $\lambda = .90$	Information	10	1.3345	42.53	-0.0209
ALL	Affine $\lambda = .90$	Periodic	20	1.3426	43.11	-0.0128
ALL	Affine $\lambda = .90$	Random	20	1.3268	42.69	-0.0286
ALL	Affine $\lambda = .90$	Confidence	20	1.3601	44.86	0.0047
ALL	Affine $\lambda = .90$	Envelope	20	1.3239	41.97	-0.0314
ALL	Affine $\lambda = .90$	Information	20	1.2947	41.28	-0.0606
Figaro	Consensus	Historical only	0	2.0320	70.83	0.0000
Figaro	Consensus	Periodic	5	2.0320	70.83	0.0000
Figaro	Consensus	Random	5	2.0320	70.83	0.0000
Figaro	Consensus	Confidence	5	2.0320	70.83	0.0000
Figaro	Consensus	Envelope	5	2.0320	70.83	0.0000
Figaro	Consensus	Information	5	2.0320	70.83	0.0000
Figaro	Consensus	Periodic	10	2.0320	70.83	0.0000

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE
Figaro	Consensus	Random	10	2.0320	70.83	0.0000
Figaro	Consensus	Confidence	10	2.0320	70.83	0.0000
Figaro	Consensus	Envelope	10	2.0320	70.83	0.0000
Figaro	Consensus	Information	10	2.0320	70.83	0.0000
Figaro	Consensus	Periodic	20	2.0320	70.83	0.0000
Figaro	Consensus	Random	20	2.0320	70.83	0.0000
Figaro	Consensus	Confidence	20	2.0320	70.83	0.0000
Figaro	Consensus	Envelope	20	2.0320	70.83	0.0000
Figaro	Consensus	Information	20	2.0320	70.83	0.0000
Figaro	Midpoint	Historical only	0	2.0884	60.06	0.0564
Figaro	Midpoint	Periodic	5	2.0877	60.22	-0.0007
Figaro	Midpoint	Random	5	2.1283	61.58	0.0399
Figaro	Midpoint	Confidence	5	2.0681	61.78	-0.0203
Figaro	Midpoint	Envelope	5	2.1703	61.39	0.0819
Figaro	Midpoint	Information	5	2.0737	60.44	-0.0148
Figaro	Midpoint	Periodic	10	2.1612	64.50	0.0727
Figaro	Midpoint	Random	10	2.1634	62.88	0.0750
Figaro	Midpoint	Confidence	10	2.0873	62.00	-0.0011
Figaro	Midpoint	Envelope	10	2.1203	61.00	0.0319
Figaro	Midpoint	Information	10	2.0733	61.17	-0.0151
Figaro	Midpoint	Periodic	20	2.2206	64.06	0.1322
Figaro	Midpoint	Random	20	2.1371	62.87	0.0487
Figaro	Midpoint	Confidence	20	2.0566	61.94	-0.0318
Figaro	Midpoint	Envelope	20	2.1180	61.61	0.0296
Figaro	Midpoint	Information	20	2.0617	64.17	-0.0267
Figaro	Projection	Historical only	0	2.0296	70.83	-0.0024
Figaro	Projection	Periodic	5	2.0269	70.83	-0.0027
Figaro	Projection	Random	5	2.0295	70.83	-0.0001
Figaro	Projection	Confidence	5	2.0257	70.78	-0.0039
Figaro	Projection	Envelope	5	2.0296	70.83	0.0000
Figaro	Projection	Information	5	2.0296	70.83	0.0000
Figaro	Projection	Periodic	10	2.0261	70.78	-0.0035
Figaro	Projection	Random	10	2.0321	70.89	0.0025
Figaro	Projection	Confidence	10	2.0195	70.50	-0.0102
Figaro	Projection	Envelope	10	2.0294	70.83	-0.0002
Figaro	Projection	Information	10	2.0296	70.83	0.0000
Figaro	Projection	Periodic	20	2.0262	70.83	-0.0034
Figaro	Projection	Random	20	2.0285	70.79	-0.0011
Figaro	Projection	Confidence	20	2.0276	70.83	-0.0020
Figaro	Projection	Envelope	20	2.0234	70.83	-0.0062
Figaro	Projection	Information	20	2.0261	70.78	-0.0035
Figaro	Offset	Historical only	0	1.9736	69.39	-0.0584
Figaro	Offset	Periodic	5	1.9619	69.28	-0.0117
Figaro	Offset	Random	5	1.9716	69.23	-0.0020
Figaro	Offset	Confidence	5	1.9474	68.83	-0.0262
Figaro	Offset	Envelope	5	1.9728	69.17	-0.0009
Figaro	Offset	Information	5	1.9720	69.33	-0.0016
Figaro	Offset	Periodic	10	1.9567	68.83	-0.0169
Figaro	Offset	Random	10	1.9687	69.16	-0.0050
Figaro	Offset	Confidence	10	1.9385	68.06	-0.0351
Figaro	Offset	Envelope	10	1.9694	69.00	-0.0042
Figaro	Offset	Information	10	1.9719	69.39	-0.0017
Figaro	Offset	Periodic	20	1.9698	69.28	-0.0038
Figaro	Offset	Random	20	1.9628	69.09	-0.0108
Figaro	Offset	Confidence	20	1.9293	67.44	-0.0444
Figaro	Offset	Envelope	20	1.9554	68.89	-0.0182
Figaro	Offset	Information	20	1.9610	69.28	-0.0126

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE
Figaro	Affine $\lambda = 1$	Historical only	0	1.7636	57.50	-0.2684
Figaro	Affine $\lambda = 1$	Periodic	5	1.7565	57.22	-0.0071
Figaro	Affine $\lambda = 1$	Random	5	1.7608	57.24	-0.0028
Figaro	Affine $\lambda = 1$	Confidence	5	1.7509	58.11	-0.0127
Figaro	Affine $\lambda = 1$	Envelope	5	1.7593	56.61	-0.0042
Figaro	Affine $\lambda = 1$	Information	5	1.7616	57.83	-0.0020
Figaro	Affine $\lambda = 1$	Periodic	10	1.7507	57.17	-0.0129
Figaro	Affine $\lambda = 1$	Random	10	1.7588	57.19	-0.0048
Figaro	Affine $\lambda = 1$	Confidence	10	1.7503	59.17	-0.0133
Figaro	Affine $\lambda = 1$	Envelope	10	1.7556	56.17	-0.0080
Figaro	Affine $\lambda = 1$	Information	10	1.7544	56.89	-0.0092
Figaro	Affine $\lambda = 1$	Periodic	20	1.7619	57.78	-0.0016
Figaro	Affine $\lambda = 1$	Random	20	1.7537	57.19	-0.0099
Figaro	Affine $\lambda = 1$	Confidence	20	1.7648	60.11	0.0012
Figaro	Affine $\lambda = 1$	Envelope	20	1.7481	56.61	-0.0155
Figaro	Affine $\lambda = 1$	Information	20	1.7351	56.28	-0.0285
Figaro	Affine $\lambda = .95$	Historical only	0	1.7636	57.50	-0.2684
Figaro	Affine $\lambda = .95$	Periodic	5	1.7565	57.22	-0.0071
Figaro	Affine $\lambda = .95$	Random	5	1.7605	57.22	-0.0031
Figaro	Affine $\lambda = .95$	Confidence	5	1.7513	58.17	-0.0123
Figaro	Affine $\lambda = .95$	Envelope	5	1.7587	56.56	-0.0049
Figaro	Affine $\lambda = .95$	Information	5	1.7620	57.83	-0.0016
Figaro	Affine $\lambda = .95$	Periodic	10	1.7500	57.28	-0.0136
Figaro	Affine $\lambda = .95$	Random	10	1.7586	57.18	-0.0049
Figaro	Affine $\lambda = .95$	Confidence	10	1.7525	59.61	-0.0111
Figaro	Affine $\lambda = .95$	Envelope	10	1.7538	55.94	-0.0098
Figaro	Affine $\lambda = .95$	Information	10	1.7551	56.94	-0.0084
Figaro	Affine $\lambda = .95$	Periodic	20	1.7645	58.44	0.0009
Figaro	Affine $\lambda = .95$	Random	20	1.7539	57.44	-0.0097
Figaro	Affine $\lambda = .95$	Confidence	20	1.7836	60.17	0.0200
Figaro	Affine $\lambda = .95$	Envelope	20	1.7462	56.50	-0.0174
Figaro	Affine $\lambda = .95$	Information	20	1.7283	56.17	-0.0353
Figaro	Affine $\lambda = .90$	Historical only	0	1.7636	57.50	-0.2684
Figaro	Affine $\lambda = .90$	Periodic	5	1.7566	57.17	-0.0070
Figaro	Affine $\lambda = .90$	Random	5	1.7602	57.18	-0.0034
Figaro	Affine $\lambda = .90$	Confidence	5	1.7520	58.44	-0.0115
Figaro	Affine $\lambda = .90$	Envelope	5	1.7579	56.61	-0.0057
Figaro	Affine $\lambda = .90$	Information	5	1.7625	57.67	-0.0011
Figaro	Affine $\lambda = .90$	Periodic	10	1.7494	57.28	-0.0142
Figaro	Affine $\lambda = .90$	Random	10	1.7588	57.36	-0.0048
Figaro	Affine $\lambda = .90$	Confidence	10	1.7563	60.00	-0.0073
Figaro	Affine $\lambda = .90$	Envelope	10	1.7520	55.94	-0.0116
Figaro	Affine $\lambda = .90$	Information	10	1.7570	56.94	-0.0065
Figaro	Affine $\lambda = .90$	Periodic	20	1.7706	59.06	0.0070
Figaro	Affine $\lambda = .90$	Random	20	1.7553	57.82	-0.0083
Figaro	Affine $\lambda = .90$	Confidence	20	1.8222	60.67	0.0586
Figaro	Affine $\lambda = .90$	Envelope	20	1.7450	56.22	-0.0186
Figaro	Affine $\lambda = .90$	Information	20	1.7239	56.11	-0.0397
FIS	Consensus	Historical only	0	0.8820	28.67	0.0000
FIS	Consensus	Periodic	5	0.8820	28.67	0.0000
FIS	Consensus	Random	5	0.8820	28.67	-0.0000
FIS	Consensus	Confidence	5	0.8820	28.67	0.0000
FIS	Consensus	Envelope	5	0.8820	28.67	0.0000
FIS	Consensus	Information	5	0.8820	28.67	0.0000
FIS	Consensus	Periodic	10	0.8820	28.67	0.0000
FIS	Consensus	Random	10	0.8820	28.67	-0.0000
FIS	Consensus	Confidence	10	0.8820	28.67	0.0000

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE
FIS	Consensus	Envelope	10	0.8820	28.67	0.0000
FIS	Consensus	Information	10	0.8820	28.67	0.0000
FIS	Consensus	Periodic	20	0.8820	28.67	0.0000
FIS	Consensus	Random	20	0.8820	28.67	-0.0000
FIS	Consensus	Confidence	20	0.8820	28.67	0.0000
FIS	Consensus	Envelope	20	0.8820	28.67	0.0000
FIS	Consensus	Information	20	0.8820	28.67	0.0000
FIS	Midpoint	Historical only	0	1.0259	34.89	0.1439
FIS	Midpoint	Periodic	5	0.9915	32.89	-0.0343
FIS	Midpoint	Random	5	1.0074	33.75	-0.0184
FIS	Midpoint	Confidence	5	0.9715	34.94	-0.0544
FIS	Midpoint	Envelope	5	1.0377	35.17	0.0119
FIS	Midpoint	Information	5	0.9634	33.50	-0.0625
FIS	Midpoint	Periodic	10	0.9908	34.50	-0.0350
FIS	Midpoint	Random	10	0.9985	33.72	-0.0273
FIS	Midpoint	Confidence	10	0.9697	35.17	-0.0562
FIS	Midpoint	Envelope	10	1.0196	33.28	-0.0062
FIS	Midpoint	Information	10	0.9275	31.06	-0.0984
FIS	Midpoint	Periodic	20	1.0129	35.89	-0.0130
FIS	Midpoint	Random	20	0.9710	33.87	-0.0548
FIS	Midpoint	Confidence	20	0.9700	36.06	-0.0559
FIS	Midpoint	Envelope	20	0.9586	32.56	-0.0672
FIS	Midpoint	Information	20	0.9141	32.44	-0.1118
FIS	Projection	Historical only	0	0.8842	28.67	0.0022
FIS	Projection	Periodic	5	0.8805	28.39	-0.0036
FIS	Projection	Random	5	0.8818	28.56	-0.0024
FIS	Projection	Confidence	5	0.8790	29.67	-0.0052
FIS	Projection	Envelope	5	0.8830	28.67	-0.0012
FIS	Projection	Information	5	0.8759	28.22	-0.0083
FIS	Projection	Periodic	10	0.8740	29.06	-0.0102
FIS	Projection	Random	10	0.8784	28.67	-0.0058
FIS	Projection	Confidence	10	0.8705	29.94	-0.0137
FIS	Projection	Envelope	10	0.8830	28.67	-0.0012
FIS	Projection	Information	10	0.8727	28.22	-0.0115
FIS	Projection	Periodic	20	0.8821	29.11	-0.0021
FIS	Projection	Random	20	0.8704	28.79	-0.0137
FIS	Projection	Confidence	20	0.8574	29.72	-0.0268
FIS	Projection	Envelope	20	0.8754	28.89	-0.0087
FIS	Projection	Information	20	0.8650	28.67	-0.0192
FIS	Offset	Historical only	0	0.9467	29.06	0.0648
FIS	Offset	Periodic	5	0.9283	28.83	-0.0184
FIS	Offset	Random	5	0.9346	28.87	-0.0121
FIS	Offset	Confidence	5	0.8992	28.28	-0.0475
FIS	Offset	Envelope	5	0.9311	28.78	-0.0156
FIS	Offset	Information	5	0.9307	28.94	-0.0160
FIS	Offset	Periodic	10	0.9099	28.28	-0.0369
FIS	Offset	Random	10	0.9215	28.73	-0.0252
FIS	Offset	Confidence	10	0.8822	27.61	-0.0646
FIS	Offset	Envelope	10	0.9265	28.50	-0.0202
FIS	Offset	Information	10	0.9190	28.89	-0.0278
FIS	Offset	Periodic	20	0.9185	28.17	-0.0283
FIS	Offset	Random	20	0.9069	28.13	-0.0398
FIS	Offset	Confidence	20	0.8694	26.89	-0.0774
FIS	Offset	Envelope	20	0.9008	28.06	-0.0459
FIS	Offset	Information	20	0.8896	28.06	-0.0571
FIS	Affine $\lambda = 1$	Historical only	0	0.9471	28.61	0.0651
FIS	Affine $\lambda = 1$	Periodic	5	0.9319	28.11	-0.0152

Scope	Estimator	Schedule	B	MAE	P_1	Δ MAE
FIS	Affine $\lambda = 1$	Random	5	0.9351	28.37	-0.0120
FIS	Affine $\lambda = 1$	Confidence	5	0.9029	28.22	-0.0442
FIS	Affine $\lambda = 1$	Envelope	5	0.9302	28.33	-0.0169
FIS	Affine $\lambda = 1$	Information	5	0.9271	28.28	-0.0200
FIS	Affine $\lambda = 1$	Periodic	10	0.9071	28.06	-0.0400
FIS	Affine $\lambda = 1$	Random	10	0.9211	28.18	-0.0260
FIS	Affine $\lambda = 1$	Confidence	10	0.8832	27.72	-0.0639
FIS	Affine $\lambda = 1$	Envelope	10	0.9273	28.11	-0.0198
FIS	Affine $\lambda = 1$	Information	10	0.9179	28.33	-0.0292
FIS	Affine $\lambda = 1$	Periodic	20	0.9179	27.89	-0.0292
FIS	Affine $\lambda = 1$	Random	20	0.9062	27.94	-0.0409
FIS	Affine $\lambda = 1$	Confidence	20	0.8709	26.89	-0.0762
FIS	Affine $\lambda = 1$	Envelope	20	0.9033	27.67	-0.0438
FIS	Affine $\lambda = 1$	Information	20	0.8828	27.67	-0.0643
FIS	Affine $\lambda = .95$	Historical only	0	0.9471	28.61	0.0651
FIS	Affine $\lambda = .95$	Periodic	5	0.9306	28.06	-0.0165
FIS	Affine $\lambda = .95$	Random	5	0.9343	28.36	-0.0128
FIS	Affine $\lambda = .95$	Confidence	5	0.9006	28.33	-0.0465
FIS	Affine $\lambda = .95$	Envelope	5	0.9290	28.33	-0.0181
FIS	Affine $\lambda = .95$	Information	5	0.9254	28.28	-0.0217
FIS	Affine $\lambda = .95$	Periodic	10	0.9037	27.83	-0.0434
FIS	Affine $\lambda = .95$	Random	10	0.9187	28.13	-0.0284
FIS	Affine $\lambda = .95$	Confidence	10	0.8790	27.44	-0.0681
FIS	Affine $\lambda = .95$	Envelope	10	0.9267	28.17	-0.0204
FIS	Affine $\lambda = .95$	Information	10	0.9152	28.28	-0.0319
FIS	Affine $\lambda = .95$	Periodic	20	0.9156	27.56	-0.0315
FIS	Affine $\lambda = .95$	Random	20	0.9018	27.79	-0.0453
FIS	Affine $\lambda = .95$	Confidence	20	0.8781	25.56	-0.0690
FIS	Affine $\lambda = .95$	Envelope	20	0.9013	27.67	-0.0458
FIS	Affine $\lambda = .95$	Information	20	0.8721	27.00	-0.0750
FIS	Affine $\lambda = .90$	Historical only	0	0.9471	28.61	0.0651
FIS	Affine $\lambda = .90$	Periodic	5	0.9291	28.00	-0.0180
FIS	Affine $\lambda = .90$	Random	5	0.9335	28.36	-0.0136
FIS	Affine $\lambda = .90$	Confidence	5	0.8983	28.44	-0.0488
FIS	Affine $\lambda = .90$	Envelope	5	0.9278	28.11	-0.0193
FIS	Affine $\lambda = .90$	Information	5	0.9234	28.28	-0.0237
FIS	Affine $\lambda = .90$	Periodic	10	0.8996	27.72	-0.0475
FIS	Affine $\lambda = .90$	Random	10	0.9161	28.08	-0.0310
FIS	Affine $\lambda = .90$	Confidence	10	0.8769	27.22	-0.0702
FIS	Affine $\lambda = .90$	Envelope	10	0.9267	28.44	-0.0204
FIS	Affine $\lambda = .90$	Information	10	0.9119	28.11	-0.0352
FIS	Affine $\lambda = .90$	Periodic	20	0.9146	27.17	-0.0325
FIS	Affine $\lambda = .90$	Random	20	0.8983	27.55	-0.0488
FIS	Affine $\lambda = .90$	Confidence	20	0.8979	29.06	-0.0492
FIS	Affine $\lambda = .90$	Envelope	20	0.9028	27.72	-0.0443
FIS	Affine $\lambda = .90$	Information	20	0.8655	26.44	-0.0816

The following files retain the complete confirmation records and full-precision presentation inputs.

File	Contents
C5_METRICS.csv	All run, configuration and subset metrics
C5_AGGREGATE.csv	Frozen seed/day aggregates
C5_PRIMARY_TABLE.csv	Primary pooled and family results
C5_PRIMARY_SESSION_TABLE.csv	Every primary session/family cell
C5_B20_AFFINE_CHANGES.csv	All fifteen affine B20 comparisons
C5_INTERVAL_TABLE.csv	Noncontainment, contradictions and assurance
C5_RESULTS.md	Scientific criteria and execution evidence

Raw noncontainment, false assurance and adverse cells remain valid scientific failures; leakage, integrity, quota or missing-group failures would invalidate evaluation, but none was observed. Reference-relative errors and descriptive later-session agreement do not establish certified uncertainty, independent-array validation or algorithmic novelty.

Declarations—draft for author approval

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Author approval. All-author approval of the manuscript, declarations and any submission has not yet been supplied.

AI-assisted work—draft disclosure for author approval

OpenAI ChatGPT and Codex assisted software, analysis/reporting code and manuscript preparation as described in the main manuscript. Confirmation of the complete tool inventory and human verification of AI-assisted work remains pending; all-author approval has not been supplied.

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